Numerical Analysis: Practice Midterm (30 marks) Solutions

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Question 1 (8 marks)**.** True or false?

1. Let $(\bullet)_2$ denote binary representation. It holds that $(0.1111)_2 + (0.0001)_2 = 1$.
2. It holds that $(1000)_2 \times (0.001)_2 = 1$.
3. It holds that
$$(0.\overline{1})_3 = \frac{1}{2}.$$
4. In base 16, all the natural numbers from 1 to 200 can be represented using 2 digits.
5. In Julia, `Float64(.1) == Float32(.1)` evaluates to `true`.
6. The spacing (in absolute value) between successive double-precision (`Float64`) floating point numbers is constant.
7. It holds that $(0.\overline{10101})_2 = (1.2345)_{10}$.
8. Machine addition $\widehat{+}$ is an associative operation. More precisely, given any three double-precision floating point numbers x, y and z, the following equality holds:
$$(x \mathbin{\widehat{+}} y) \mathbin{\widehat{+}} z = x \mathbin{\widehat{+}} (y \mathbin{\widehat{+}} z).$$
9. The machine epsilon is the smallest strictly positive number that can be represented in a floating point format.
10. Let ε denote the machine epsilon for the double-precision format. Let also $\widehat{+}$ and $\widehat{/}$ denote respectively the machine addition and the machine division operators for the double-precision format. It holds that $1 \mathbin{\widehat{+}} (\varepsilon \mathbin{\widehat{/}} 64) = 1$ and that $\varepsilon \mathbin{\widehat{/}} 64 \neq 0$.
11. Assume that $x \in \mathbf{R}$ belongs to the double-precision floating point format (that is, assume that $x \in \mathbf{F}_{64}$). Then $-x \in \mathbf{F}_{64}$.

A correct (resp. incorrect) answer leads to +1 mark (resp. -1 mark).

Solution. 1. **True**: $(0.1111)_2 + (0.0001)_2 = \frac{15}{16} + \frac{1}{16} = 1$.

2. **True**: $(1000)_2 = 8$ and $(0.001)_2 = \frac{1}{8}$.
3. **True**: $(0.\overline{1})_3 = \sum_{k=1}^{\infty} 3^{-k} = \frac{1/3}{1-1/3} = \frac{1}{2}$.
4. **True**: with two hexadecimal digits, all the numbers up to $(\mathrm{FF})_{16} = 255$ can be represented.
5. **False**: $0.1 = (0.0\overline{0011})_2$ is not exactly representable, and its roundings to single and double precision differ.
6. **False**: the spacing between successive numbers in $[2^e, 2^{e+1})$ is 2^{e-52}, which depends on e.
7. **False**: the left-hand side is less than 1, whereas the right-hand side is greater than 1.
8. **False**: for example, $(1 \mathbin{\widehat{+}} \varepsilon/2) \mathbin{\widehat{+}} \varepsilon/2 = 1$, whereas $1 \mathbin{\widehat{+}} (\varepsilon/2 \mathbin{\widehat{+}} \varepsilon/2) = 1 + \varepsilon$.
9. **False**: the machine epsilon is the distance between 1 and the next floating point number; much smaller numbers are representable, e.g. 2^{-1022}.
10. **True**: $\varepsilon \mathbin{\widehat{/}} 64 = 2^{-58}$ is exactly representable, and in particular nonzero. Since 2^{-58} is much smaller than half the spacing ε between 1 and the next floating point number, $1 \mathbin{\widehat{+}} 2^{-58} = 1$.
11. **True**: the sign is stored in a separate bit, so the set $\mathbf{F}_{64}$ is symmetric with respect to 0.

△

Question 2 (8 marks)**.** Assume that $\mathsf{A} \in \mathbf{R}^{n\times n}$ is an invertible matrix and that $\boldsymbol{b} \in \mathbf{R}^n$ and $\boldsymbol{\beta} \in \mathbf{R}^n$ are two nonzero vectors in $\mathbf{R}^n$. We denote by $\boldsymbol{x}$ and $\boldsymbol{\xi}$ the solutions to the linear equations $\mathsf{A}\boldsymbol{x} = \boldsymbol{b}$ and $\mathsf{A}\boldsymbol{\xi} = \boldsymbol{\beta}$, respectively. Show that

$$\frac{\|\boldsymbol{x} - \boldsymbol{\xi}\|}{\|\boldsymbol{x}\|} \leqslant \|\mathsf{A}\|\|\mathsf{A}^{-1}\|\frac{\|\boldsymbol{b} - \boldsymbol{\beta}\|}{\|\boldsymbol{b}\|}.$$

Here $\|\bullet\|$ denotes both the Euclidean vector norm and the induced matrix norm.
Bonus question (1 mark): Let $\kappa := \|\mathsf{A}\|\|\mathsf{A}^{-1}\|$. Prove that $\kappa \geqslant 1$.

Solution. By linearity, $\mathsf{A}(\boldsymbol{x} - \boldsymbol{\xi}) = \boldsymbol{b} - \boldsymbol{\beta}$, and so

$$\|\boldsymbol{x} - \boldsymbol{\xi}\| = \|\mathsf{A}^{-1}(\boldsymbol{b} - \boldsymbol{\beta})\| \leqslant \|\mathsf{A}^{-1}\|\|\boldsymbol{b} - \boldsymbol{\beta}\|.$$

On the other hand, $\|\boldsymbol{b}\| = \|\mathsf{A}\boldsymbol{x}\| \leqslant \|\mathsf{A}\|\|\boldsymbol{x}\|$, so $\frac{1}{\|\boldsymbol{x}\|} \leqslant \frac{\|\mathsf{A}\|}{\|\boldsymbol{b}\|}$ (note that $\boldsymbol{x} \neq \mathbf{0}$ since $\boldsymbol{b} \neq \mathbf{0}$). Multiplying these two inequalities gives the result.

Bonus: since the induced norm of the identity matrix is 1 and the induced norm is submultiplicative,

$$1 = \|\mathsf{I}\| = \|\mathsf{A}\mathsf{A}^{-1}\| \leqslant \|\mathsf{A}\|\|\mathsf{A}^{-1}\| = \kappa.$$

△

Question 3 (8 marks)**.** Let $\mathsf{A} \in \mathbf{R}^{n\times n}$ be a symmetric positive definite matrix and let $\boldsymbol{b} \in \mathbf{R}^n$. The steepest descent algorithm for solving $\mathsf{A}\boldsymbol{x} = \boldsymbol{b}$ is given hereafter:

Pick $\varepsilon > 0$ and initial $\boldsymbol{x}$
$\boldsymbol{r} \leftarrow \mathsf{A}\boldsymbol{x} - \boldsymbol{b}$
while $\|\boldsymbol{r}\| \geqslant \varepsilon\|\boldsymbol{b}\|$ **do**
  $\omega \leftarrow \boldsymbol{r}^T\boldsymbol{r}/\boldsymbol{r}^T\mathsf{A}\boldsymbol{r}$
  $\boldsymbol{x} \leftarrow \boldsymbol{x} - \omega\boldsymbol{r}$
  $\boldsymbol{r} \leftarrow \mathsf{A}\boldsymbol{x} - \boldsymbol{b}$
end while

- Why is this method called the *steepest descent* algorithm? (1 mark)
- How many floating point operations does an iteration of this algorithm require? (5 marks)
- Are the following statements true of false? (2 marks)

  1. There exists a unique solution $\boldsymbol{x}_*$ to the linear system $\mathsf{A}\boldsymbol{x} = \boldsymbol{b}$.
  2. The iterates converge to $\boldsymbol{x}_*$ in at most n iterations.
  3. We consider the following modification of the algorithm:

     Pick $\varepsilon > 0$, $\omega > 0$ and initial $\boldsymbol{x}$
     $\boldsymbol{r} \leftarrow \mathsf{A}\boldsymbol{x} - \boldsymbol{b}$
     while $\|\boldsymbol{r}\| \geqslant \varepsilon\|\boldsymbol{b}\|$ **do**
       $\boldsymbol{x} \leftarrow \boldsymbol{x} - \omega\boldsymbol{r}$
       $\boldsymbol{r} \leftarrow \mathsf{A}\boldsymbol{x} - \boldsymbol{b}$
     end while

     If ω is sufficiently small, then this algorithm converges.
  4. Here we no longer assume that A is positive definite. Instead, we consider that

$$\mathsf{A} = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}.$$

     In this case, the steepest descent algorithm is convergent for any initial $\boldsymbol{x}$.

Solution.

- Solving $\mathsf{A}\boldsymbol{x} = \boldsymbol{b}$ is equivalent to minimizing $f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\mathsf{A}\boldsymbol{x} - \boldsymbol{b}^T\boldsymbol{x}$, whose gradient is $\nabla f(\boldsymbol{x}) = \mathsf{A}\boldsymbol{x} - \boldsymbol{b} = \boldsymbol{r}$. At each iteration, the algorithm moves in the direction $-\boldsymbol{r}$, which is the direction of steepest descent of f, and the step ω is the one that minimizes $f(\boldsymbol{x} - \omega\boldsymbol{r})$ along this direction.

- We count the operations in one iteration of the loop:
  - $\mathsf{A}\boldsymbol{r}$: n^2 multiplications and $n(n-1)$ additions, i.e. $2n^2 - n$ flops;
  - $\boldsymbol{r}^T\boldsymbol{r}$ and $\boldsymbol{r}^T(\mathsf{A}\boldsymbol{r})$: $2n - 1$ flops each;
  - the division to compute ω: 1 flop;
  - $\boldsymbol{x} - \omega\boldsymbol{r}$: $2n$ flops;
  - $\mathsf{A}\boldsymbol{x} - \boldsymbol{b}$: $2n^2 - n + n = 2n^2$ flops;
  - the stopping criterion: $\|\boldsymbol{r}\|$ requires $2n - 1$ flops and a square root ($\varepsilon\|\boldsymbol{b}\|$ can be computed once and for all).

  The total is approximately $4n^2 + 7n$ flops per iteration. (The cost can be reduced to $2n^2 + \mathcal{O}(n)$ by updating the residual as $\boldsymbol{r} \leftarrow \boldsymbol{r} - \omega\mathsf{A}\boldsymbol{r}$, reusing the product $\mathsf{A}\boldsymbol{r}$.)

- 1. **True**: a symmetric positive definite matrix is invertible.
  2. **False**: the steepest descent method does not terminate after finitely many iterations in general; convergence in at most n iterations (in exact arithmetic) is a property of the conjugate gradient method. For the steepest descent method, the error decreases geometrically, with a rate depending on the condition number of A.
  3. **True**: this is Richardson's method, whose iteration matrix is $\mathsf{I} - \omega\mathsf{A}$. Its eigenvalues $1 - \omega\lambda_i$ belong to $(-1, 1)$ if $0 < \omega < \frac{2}{\lambda_{\max}}$, in which case the method converges.
  4. **True**: let $\mathsf{B} = -\mathsf{A}$, which is symmetric positive definite. Then $\boldsymbol{r} = \mathsf{A}\boldsymbol{x} - \boldsymbol{b} = -(\mathsf{B}\boldsymbol{x} + \boldsymbol{b})$ and $\omega = -\frac{\boldsymbol{r}^T\boldsymbol{r}}{\boldsymbol{r}^T\mathsf{B}\boldsymbol{r}}$, so the update $\boldsymbol{x} - \omega\boldsymbol{r}$ coincides with one iteration of the steepest descent method for the system $\mathsf{B}\boldsymbol{x} = -\boldsymbol{b}$, whose solution is the same as that of $\mathsf{A}\boldsymbol{x} = \boldsymbol{b}$. Since B is symmetric positive definite, the method converges for any initial $\boldsymbol{x}$.

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Question 4 (6 marks)**.** We proved in class the quadratic convergence of the Newton–Raphson method for a smooth function with a simple root. The aim of this exercise is to study the convergence of the method in the case of a function with a double root. To this end, we consider the simple one-dimensional equation

$$f(x) := (x - 1)^2 = 0. \tag{1}$$

1. Write down one iteration of the Newton–Raphson method for (1) in the form:

$$x_{k+1} = F(x_k).$$

2. Let $e_k = x_k - x_*$, where x_* is the exact solution to (1). Find a recurrence relation for the error and, assuming that the initial guess is $x_0 = 2$, write down an explicit expression for e_k.

3. What is the order of convergence of the method in this case?

4. **Bonus question** (1 mark): Repeat the previous exercises for the equation $(x-1)^3 = 0$. What is the order of convergence in this case, and what is the rate of convergence?

Solution. 1. Since $f'(x) = 2(x-1)$, the Newton–Raphson iteration reads, for $x_k \neq 1$,

$$x_{k+1} = x_k - \frac{(x_k-1)^2}{2(x_k-1)} = x_k - \frac{x_k-1}{2} = \frac{x_k+1}{2} =: F(x_k).$$

2. The exact solution is $x_* = 1$, and $e_{k+1} = x_{k+1} - 1 = \frac{x_k-1}{2} = \frac{e_k}{2}$. For $x_0 = 2$, we have $e_0 = 1$ and so $e_k = 2^{-k}$.

3. The convergence is only linear (order 1), with rate $\frac{1}{2}$. The quadratic convergence is lost because the root is not simple.

4. For $f(x) = (x-1)^3$, we have $f'(x) = 3(x-1)^2$, and

$$x_{k+1} = x_k - \frac{x_k-1}{3}, \qquad e_{k+1} = \frac{2}{3}e_k, \qquad e_k = \left(\frac{2}{3}\right)^k \text{ for } x_0 = 2.$$

The convergence is again linear, with rate $\frac{2}{3}$. (More generally, for a root of multiplicity m, the rate is $1 - \frac{1}{m}$.)

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